

Wind Energy Generation and Storage System (Project Report)

1. Aim of the Project

The goal of this project is to build a small, working model that captures **wind energy** (kinetic energy), converts it into **electrical energy**, stores that energy in a rechargeable battery, and uses the stored charge to light up LEDs.

2. Components Used (The 'Ingredients')

Component	What it Is	Function in the Project
Wind Turbine Blades/Rotor	The fan-like structure attached to the motor shaft.	Catches the wind and starts spinning. This converts wind's speed (kinetic energy) into rotational energy.
Dynamo (Generator)	Small 5V DC Motor	When the blades spin this motor, it works in reverse, generating electricity (DC voltage). This is the key energy conversion step.
Storage Battery	18650 Lithium-Ion Battery	A high-power, rechargeable battery that stores the electricity generated by the dynamo.
Safety and Control Unit	BMS (Battery Management System)	This is a crucial safety circuit. It acts as a guard, ensuring the battery doesn't get dangerously overcharged or fully drained (over-discharged).
Indicator/Load	Generic 3V LEDs	These light bulbs show us that the project is working and that the battery is holding a charge.
Wiring/Connectors	Wires used to connect all the components.	Transfers electrical power between the different parts of the system.

3. The Science: How the System Works

The entire process involves a chain of energy conversion:

1. **Kinetic Energy to Mechanical Energy:** Wind moves the turbine blades.
2. **Mechanical Energy to Electrical Energy:** The spinning blades turn the shaft of the 5V DC Motor. Inside the motor, magnets and coils move relative to each other, creating an electric current (this is the principle of **electromagnetic induction**).
3. **Storage and Regulation:** The raw electricity from the motor is often unstable. It is sent to the **BMS**.
 - The **BMS** cleans the power and ensures the 18650 battery receives the correct

- voltage and current to charge safely.
 - The energy is now stored as **Chemical Potential Energy** inside the Lithium-Ion battery.
4. **Electrical Output:** When we switch on the load, the stored energy flows from the battery and lights up the 3V LEDs. Because the 18650 battery's voltage (up to 4.2V) is slightly higher than the LED's requirement (3V), the LED is functioning as a visual indicator of the stored charge.

4. Why is the BMS (Battery Management System) Important?

The 18650 Lithium-Ion battery is powerful but sensitive. If it is charged too much (overcharged) or drained too much (over-discharged), it can be permanently damaged or, in severe cases, become unsafe.

The **BMS** protects the battery by:

- **Overcharge Protection:** It stops the charging current when the battery reaches its full, safe voltage (around 4.2V per cell).
- **Over-discharge Protection:** It cuts off the power supply to the LEDs or other loads when the battery voltage drops too low, saving the battery from damage.
- **Short-Circuit Protection:** It quickly cuts power if a wire connects incorrectly.

5. Final Output and Conclusion

Our project successfully demonstrates the entire cycle of sustainable energy: **Harvesting** (wind turbine), **Converting** (DC motor dynamo), **Storing** (18650 battery via BMS), and **Using** (LEDs). This small model illustrates the same principles used in large-scale commercial wind farms.